

Review gate R.2: the decisions

2026-08-29, run at tier 1 in an overnight batch session, per the card in docs/czb_work_orders.md. The gate's five questions are answered in the card's priority order. Everything quantitative below comes from committed scripts with tracked outputs; the two decisive analyses were pre-registered (models, folds, and decision rules committed before the runs — commit e9203c6). Where a decision is a scope call that belongs to Jonas, it is routed to him as a query rather than taken here.

1 Is the preference field the right scalar? No — ruled against, pre-registered

The comparison the card put first (replication/czb/cutin2_field_vs_gap.py → out/cutin2_field_vs_gap.md): the existing field, computed on the second cut-in study's traces by the exact study-1 code path ($k = 12$, continuous forms, shown-window covariates), against a gap threshold of identical model form (lapse + probit, three parameters each), on identical leave-one-starting-TTC-out folds, decision rule stated before the run.

model (best scale)	held-out wRMSE	Spearman
gap (log)	0.1522	+0.841
TTC (log)	0.1679	+0.805
required deceleration (log)	0.2888	-0.158
field	0.3471	-0.424
chance (train mean)	0.3202	—
sampling-noise floor	0.1176	—

The pre-registered rule fired at dRMSE = +0.195: **the field is ruled against on its own scenario** — it scores worse than chance on grouped folds while a three-parameter gap threshold approaches the sampling-noise floor. The verdict is robust to excluding the 20 non-modal attention-check participants (+0.202) and to a post-hoc repair of a trace-noise handicap discovered after the registered run (the 30 Hz ego-speed dither puts a ~1 000-unit control-effort floor under every field covariate; with the ego speed smoothed the field scores 0.3564 and the verdict is unchanged). Within matched-TTC rows the smoothed field's correlation with the response is negative in 14 of 24 rows (mean $p = -0.10$) [corrected 2026-09-02: this sentence originally read "orders only 14 of 24 rows in the observed direction"; the observed direction for a criticality measure is positive, so the field orders 10 of 24 rows like the data and is anti-ordered in 14 — docs/r2_pipeline_review.md §4] where the data order 24 of 24 (gap $p = -0.887$) — and the sign derivation below shows that inconsistency is structural, not noise.

Two consequences, both now in force. No document may describe the field as validated against human data on the longitudinal dimension (blocker B2.Q4's wording rule, now backed by the analysis it was waiting for). And per the card's own pre-commitment, the project's claim is restated around what survives (§4).

[Added 2026-09-02, after an adversarial review of this pipeline (docs/r2_pipeline_review.md; diagnostic replication/czb/cutin2_lane_gate_diagnostic.py → out/cutin2_lane_gate_diagnostic.md). The verdict reproduces by an independent fitting route (field 0.3508, gap 0.1531; cell-bootstrap interval on the difference +0.174 to +0.222). Its loss decomposes into two identifiable mechanisms: about 44% is the lane-entry gate — a project construction — which suppresses the deficit in the slow-lane-change cells at short TTC, exactly where participants respond most (their response is flat across lane-change duration, the gate is not); the remainder is the released counterfactual magnitude, which within matched-TTC rows orders cells by the vehicles' absolute speeds rather than by gap. Nothing in the verdict changes; its scope statement should say that both the project's gate and the published term's counterfactual fail, each for a stated reason.]

2 The C1 inversion: a covariate defect, found, fixed, and reinterpreted

Blocker B2.Q1 is closed (replication/czb/c1_covariate_defect.py → out/c1_covariate_defect.md). The inversion was not in the data or the preference function but in the covariate window: the C1 lookup included the manoeuvre-onset frame, where the lane-entry projection — whose lever arm is the longitudinal TTC — turns at most 6 cm of lateral motion into near-full predicted overlap at the largest gap ($p_{\text{lane}} 0.000 / 0.318 / 0.997$ for gaps 10.5 / 16.0 / 21.5 m); and the running maxima accumulated from the trace start, ~5–16 s before the shown clip, letting a startup blip floor every 1.5 m overtake cell at 230.9 units. The fix matches the covariate window to the shown clip (RANDOM_CLIP_LEAD_S, C1_COV_END_S in comfortzone.czb_data, with property tests). **Decision: C1 stays in the fit**

with the corrected (~ 1 -unit) covariates and identifies the lapse, exactly as the model assumes.

The second study then revises the interpretation I gave the residual C1 behaviour when raising B2.Q5. Study 1's three C1 cells are ordered by gap (0.122 / 0.070 / 0.052), which I read as an ordinary proximity response the lane-gated field cannot express. The second study's 90 CP1 cells span gaps of 3.9–82 m and are **flat** (rates 0.000–0.115, $p(\text{gap}) = -0.04$): drivers do not, in fact, respond to adjacent-lane longitudinal proximity before encroachment. The lane-gated flat floor is *supported* at pre-encroachment, and the study-1 C1 gradient is better read as exposure-driven anticipation — study 1 shows 3 conditions ~ 24 times each (learnable), study 2 spreads exposure over 72 distinct clips, and its own repeat manipulation shows the learning directly ($P(\text{intervene}) 0.547 \rightarrow 0.575$ on the second showing of an identical clip). This is R.1.Q3's account, now with the mechanism's direct test behind it.

3 The lateral term: not built, and the sign derivation that decided it

The gate's original question — re-derive Σ or adopt the Driver's Risk Field — is answered *neither*, for two reasons that arrived in order.

First, R.2.Q1's instruction was to derive the sign of the distance effect under the expected-deficit proposal before writing model code. Done numerically on the actual preference function (replication/czb/expected_deficit_sign.py $\rightarrow$ out/expected_deficit_sign.md), over the second study's own delta-velocity range: the point field's within-row direction at matched TTC is **inconsistent** — $p(\text{gap}, d) = +1.00$ at TTC 2–3 s, -1.00 at TTC 5–7 s, through the counterfactual-residual term — and distance-scaled perceptual noise shifts $E[d]$ by at most $\sim 10\%$ with inconsistent sign. The observed effect is first-order and monotone (24 of 24 rows). The expected-deficit mechanism therefore **fails as the explanation of the distance–time anomaly**, per the note's own stop rule; it remains directionally right only for the lateral-clearance problem.

Second, with §1's verdict, a lateral extension would graft a comfort term onto a longitudinal core the data have rejected. The pre-registered LTAP and overtake tests of the expectation proposal (docs/lateral_and_uncertainty_note.md §6 steps 2–3) are therefore **not run**. Kolekar et al.'s DRF is re-positioned: no longer a donor of a lateral term for our field, but a member of the comparator class for the restated program (§4) — it is an independently validated instance of "keep one scalar below a threshold", which is the claim that survives.

4 What survives, and what the headline becomes

The card asked the gate to say this explicitly, so: **the project's headline result is the trait claim, not the field**. What stands, each with its evidence:

- 1 **A driver carries one comfort-zone level across scenarios.** Per-driver criticality-adjusted propensity correlates $+0.50$ to $+0.74$ across all six scenario pairs, 0.53 – 0.78 of the reliability ceiling, mean ~ 0.69 — measured with no field at all (out/cross_scenario_consistency.md). About a third of reliable per-driver variance is scenario-specific, which caps every transfer test.
- 2 **The boundary distribution is well estimated on its own scale, and it is stable.** A.2.v2 regenerated the percentile table under $k = 12$ + the covariate fix: the 50th/80th/95th percentiles moved $+0.9\%$ / $+0.2\%$ / -0.5% (out/stage1_summary.md). The lapse–threshold correlation is partly a real trait (fitted $p = -0.717$, SE 0.128, against an estimator-artifact baseline of -0.27) and does not move the deliverable.
- 3 **Transfer works once the per-scenario lapse is freed** (re-run under the new spec this session; out/transfer_overtake_summary.md).
- 4 **The best within-scenario criticality axis on the decisive dataset is the longitudinal gap** (log scale), with TTC close behind and required deceleration nowhere — which is a statement about *any* demand-based axis, not only ours.

The restated claim, recommended wording: *each driver carries a scalar comfort-zone threshold that is substantially shared across scenarios; the right criticality axis is an open empirical question on which simple scene scalars currently beat the active-inference preference field, whose kinematic content is ruled out as fitted*. The elliptical-joint / per-scenario 2D comparator program moves from fallback to primary for the axis question — this is the fallback position the project recorded in advance, now triggered by its stated condition.

Program consequences routed to Jonas (queries R.2.Q4–Q6 in the register): whether B.2/B.3 stay field constructions or become comparator-class constructions; how the assessment document and any manuscript position the active-inference contribution (the framing, the calibration pipeline, and the falsification itself are all real contributions);

and whether the handbook's CZB chapters wait for his Word review or get a correcting note now.

5 The A.3 verdict's wording: narrowed

Done in place (docs/active_inference_for_czb_assessment.md, dated note): the FAIL stands for the accumulator we specified — deficit-driven drift, non-leaky, fixed bound, on this repeated-exposure stimulus set — and licenses no claim about evidence accumulation as a family, whose standard traffic architectures (looming or TTC drift, leaky accumulation, collapsing bounds; Bontje et al., 2026) were never tested.

6 The R.1 queries the gate was asked to settle

- **R.1.Q1** (percentile quoting): adopted as a convention in A.2.v2 — every percentile is quoted as value, CI, and specification (bias variant, k, covariate window); the summary header names the full spec.
- **R.1.Q2** (lapse–threshold correlation): settled by the correlated-effects fit — $\rho = -0.717$ (SE 0.128) against the -0.27 artifact baseline; partly real, does not move the 80th percentile beyond the pre-stated rule; hierarchical stays primary.
- **R.1.Q3** (anticipation): turned into a concrete estimator, recorded in docs/czb_validation_roadmap.md §4c item 5 — a joint study+NDS fit sharing the boundary population, with a criticality-dependent paradigm offset $\delta(x) = \delta_0 + \delta_1(x - x^*)$ identified by the slope difference; with the caution that δ_1 absorbs paradigm *and* field misspecification, which the comparison cannot separate.

References

- Bontje, F., van Waveren, F., van Maanen, L., Nallapu, B., Markkula, G., & Zgonnikov, A. (2026). *Knowing when to move: Evidence accumulation models of human behavior in traffic* (arXiv:2606.00727). arXiv. <https://arxiv.org/abs/2606.00727>
- Kolekar, S., de Winter, J., & Abbink, D. (2020). Human-like driving behaviour emerges from a risk-based driver model. *Nature Communications*, 11, Article 4850. <https://doi.org/10.1038/s41467-020-18353-4>